

# Muhammad Nabeed Haider

## Computer Science Undergraduate

Islamabad, Pakistan

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## PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience developing Python applications, backend services, and AI-powered software. Built projects using Python, FastAPI, LangGraph, SQL, and modern development tools while following modular software design principles. Seeking a Junior Python Developer role to build reliable software and continue growing as a backend engineer.

## EDUCATION

### Bachelor of Science in Computer Science

2026

FAST National University of Computer and Emerging Sciences, Islamabad

- CGPA: 3.56 / 4.00
- Relevant Coursework: Generative AI, Machine Learning, MLOps, Natural Language Processing, Statistical Modelling, Cloud Computing, Information Security

## EXPERIENCE

### AI Lab Demonstrator

Spring 2026

FAST National University of Computer and Emerging Sciences

- Conducted weekly Artificial Intelligence laboratory sessions for approximately 50 undergraduate students.
- Provided guidance on Python programming, search algorithms, machine learning fundamentals, and debugging techniques.
- Assisted with grading programming assignments, evaluating lab performance, and supporting students during practical sessions.
- Collaborated with the course instructor to maintain a consistent and engaging laboratory environment.

## PROJECTS

### GymBuddy

2026

*Python • FastAPI • Ollama*

- Developed the backend of a fitness management application using Python and FastAPI with a modular REST API architecture.
- Designed APIs for workout planning, exercise management, and user data handling.
- Collaborated within a development team using Git and version control practices to implement and integrate application features.
- Focused on writing maintainable backend code with clear project structure and scalable API design principles.

### SCAR

2026

*Python • LangGraph • FastAPI • Ollama*

- Designed a modular multi-agent AI consulting platform using LangGraph and Python.
- Implemented specialized AI agents capable of task delegation and collaborative reasoning using locally hosted large language models through Ollama.

- Developed a scalable architecture emphasizing modularity, extensibility, and maintainability to support future enterprise AI workflows.
- Integrated FastAPI to expose agent capabilities through a RESTful backend suitable for production deployment.

### **Production MLOps Pipeline**

2026

*Python • FastAPI • Docker • GitHub Actions*

- Built a production-ready machine learning deployment pipeline using FastAPI and Docker with a modular project architecture.
- Implemented automated CI workflows with GitHub Actions to validate code quality, execute tests, and streamline deployment.
- Integrated Prometheus metrics and Grafana dashboards to monitor application health, inference performance, and operational metrics.
- Designed a scalable deployment workflow following MLOps best practices for reproducibility, observability, and maintainability.

### **FakeLess**

2025–2026

*Python • PyTorch • SpeechBrain • HiFi-GAN*

- Developed an audio deepfake prevention system that learns imperceptible perturbations to reduce the effectiveness of neural voice cloning models while preserving speech quality.
- Designed and trained deep learning models using PyTorch, leveraging SpeechBrain speaker embeddings and HiFi-GAN neural vocoders within the training pipeline.
- Prepared and processed the LibriSpeech dataset, implementing preprocessing, feature extraction, and automated training workflows for large-scale experimentation.
- Evaluated the system using speaker similarity and speech quality metrics to balance robustness against voice cloning with perceptual audio quality.

## **TECHNICAL SKILLS**

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**Languages:** Python, Java, SQL, C/C++

**Backend:** FastAPI, REST APIs

**Databases:** SQLite, MongoDB, PostgreSQL

**Tools:** Git, GitHub, Docker, Linux, VS Code

**Concepts:** Object-Oriented Programming, Data Structures, Algorithms, Software Engineering

## **ACHIEVEMENTS**

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- AI Lab Demonstrator for the Artificial Intelligence course at FAST-NUCES.
- Developed multiple end-to-end AI and machine learning projects spanning Generative AI, MLOps, and backend systems.
- Hands-on experience with production-oriented machine learning workflows, CI/CD, Docker, monitoring, and LLM-based applications.